

**Development of Methods to Investigate Psilocybin-Induced Neural Plasticity in
Planarian Flatworms: Optimizing Immunohistochemistry Protocols and
Preliminary Behavioural Analysis**

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Background: Psilocybin shows promise in treating various psychiatric conditions, with evidence suggesting its therapeutic effects are mediated through enhanced neural plasticity. While rodent models have provided valuable insights into these mechanisms, their use is limited by cost and time constraints. Planarian flatworms, with their simple yet homologous nervous system, complex behavioural profile, and ease of manipulation, may serve as a complementary model organism for studying psilocybin's effects on neural plasticity and behaviour.

Methods: We adapted an immunohistochemistry protocol for anti-phospho-histone-H3 (PH3) staining in planarians through multiple iterations. We then conducted a pilot dose-response study examining planarian locomotion after exposure to various psilocybin concentrations (1-40 μ M). We also attempted to assess cell proliferation in psilocybin-exposed planarians (20 μ M) using our optimized PH3 staining protocol.

Results: Protocol optimisation identified several key factors that affect staining quality, including mucus removal time, bleaching duration, and antibody concentration. Results varied, and suggest further optimisation is required. Preliminary behavioural data suggested dose-dependent effects of psilocybin on planarian locomotion. However, cell proliferation analysis in psilocybin-exposed planarians was inconclusive due to inconsistent staining results.

Conclusions: While we saw some success in adapting a PH3 staining protocol, further optimization is needed for reliable quantification of cell proliferation in psilocybin-exposed planarians. Future research should focus on systematic protocol refinement, species identification of the planarians, and development of additional assays for examining psilocybin's effects on neural plasticity and behaviour in this model organism.

Introduction

Psilocybin (O-phosphoryl-4-hydroxy-N, N-dimethyltryptamine) is a naturally occurring hallucinogenic drug which acutely induces perceptual changes, ego dissolution, and a range of easily alterable moods (Griffiths et al., 2006). Numerous randomised control trials indicate psilocybin as an effective treatment for depression (Haikaizian et al, 2023) These effects are seen in both treatment resistant depression (Goodwin et al., 2022), and major depressive disorder (Davis et al., 2021). Research also suggests psilocybin is an effective treatment for addiction (Johnson et al., 2014), (Bogenschutz et al., 2015) obsessive compulsive disorder (Moreno et al., 2006), and end of life anxiety (Ross et al., 2016). Psilocybin has also been seen to increase life satisfaction and sense of well-being in healthy volunteers (Nicholas et al., 2018). A single high dose can induce these effects, which are both acutely induced and persistent. Patients frequently experience initial side effects, but the burden of those side effects is generally low (Haikaizian et al., 2023). The results are also often notably larger than behavioural interventions without psychopharmacology, or with other psychiatric drugs. Thus, there is rapidly growing interest in psilocybin's therapeutic capabilities.

Rodent models are frequently used to investigate the effects of psilocybin, demonstrating outcomes of interest. In both mice and rats, psilocybin has been shown to induce anti-depressant like effects (Zhao et al., 2024), (Hesselgrave et al., 2021), (Hibicke et al., 2020). Studies have also found it to attenuate learned-helplessness behaviour (Shao et al., 2021), and promote fear extinction (Du et al., 2023), (Catlow et al., 2013). However, Jefsen et al. (2019) did not find anti-depressant effects in Flinders Line rats, and Meinhardt et al. (2020) found no long-term reductions in alcohol relapse.

Lasting structural changes are observed in the brains of rodents after psilocybin administration. These include synaptogenesis, spinogenesis, and neuritogenesis (Ly et al., 2020) in the prefrontal cortex (PFC), as well as neurogenesis in the hippocampus (Du et al., 2023). There are also changes to the three-dimensional shape of dendritic spines in the PFC, which was found to be associated with increased excitability (Shao et al., 2021). In cortical regions, alterations in calcium signalling and modifications to synaptic integration have also been observed (Savalia et al., 2021).

In humans, acute changes in functional neural activity have been observed (Banks et al., 2021). These include altered connectivity between the amygdala and key brain regions such as the PFC (Banks et al., 2021), a disruption of connectivity within the default mode network (Banks et al., 2021), (Siegel et al., 2024), and increases in between-network connectivity (Siegel et al., 2024). The changes in functional connectivity have been found to last as long as three weeks (Siegel et al., 2024), but the full extent to which these effects are persistent is unclear.

Both the structural and functional changes that psilocybin induces are thought to underlie the therapeutic benefits of the drug. Thus, there is significant body of research investigating the underlying mechanisms by which these changes are induced.

The 5-HT_{2A} receptor (5-HT_{2A}R) is the most commonly implicated mechanism in the literature. Psilocybin is a partial agonist to the 5-HT_{2A}R. Ketanserin, a 5-HT_{2A} antagonist, blocks both the neurite growth (Ly et al., 2020) and the subjective psychotic effects induced by psilocybin (Vollenweider et al., 1998). The receptor occupancy of the 5-HT_{2A} predicts the strength of the subjective effects of psilocybin (dose-dependent relationship) (Stenbaek et al., 2021, Madsen et al., 2019). Thus, 5-HT_{2A}Rs are thought to play a significant role in both the neurological changes post psilocybin exposure, and the subjective perceptual changes that are acutely induced.

5-HT_{2A}Rs are particularly dense on layer 5 pyramidal neurons, and it is here that they are particularly implicated in psilocybin's effects. Activation of these receptors in the rat frontal cortex leads to acute membrane depolarization (Savalia et al., 2021). This induces a complex signalling pathway. Extracellular glutamate is released in the frontal cortex (Wojtas et al., 2022), (Mason et al., 2020). Glutamate release activates metabotropic glutamate receptors, which are involved in the regulation of neural

plasticity (Mukherjee et al., 2013). Group II mGlu activation is critically required for persistent LTD induction and maintenance. (Mukherjee et al., 2013). Agonists for mGluR2 suppresses psychedelic responses (Banks et al., 2021). Furthermore, in mGluR2 knockout mice, some of the behavioural and cellular effects of psychedelic administration are not seen (Moreno et al., 2011). This is possibly due to the ability of 5-HT_{2A} and mGluR2 to form heterodimeric complexes, creating a distinct signalling profile and leading to a modulation of downstream effects (Banks et al., 2021). This extracellular glutamate also results in the stimulation of ionotropic glutamate receptors (Ly2020). When DQNX (an AMPA receptor blocker) is used, psychedelic induced dendritogenesis is blocked during both the initiation and growth phase (Ly et al., 2020), (Jourdi2009). This implies that AMPARs play a role in the maintenance of neural plasticity after psilocybin exposure, as well as the induction. The stimulation of these receptors may contribute to neural plasticity through secretion of brain-derived neurotrophic factor (BDNF) (Jourdi et al., 2009), (Lisman et al., 2012).

TrkB, a high affinity receptor for BDNF, also appears to be important for the structural changes induced by psilocybin (Ly et al., 2018). When cortical cultures are treated with selective antagonists of the TrkB receptor, psychedelics and BDNF fail to stimulate neuritogenesis and spinogenesis (Ly et al., 2018). Once BDNF has bound to TrkB, TrkB activates mTOR. This induces neuritogenesis (Ly et al., 2018). Rapamycin, an mTOR inhibitor, blocks the increases in neuritogenesis seen after psychedelic administration (Ly et al., 2018). Like AMPA receptors, sustained mTOR activation appears to be key for enhanced neuroplasticity (Ly et al., 2018).

Thus, many key neurological factors for the neural plasticity induced by psilocybin exposure have been identified. However, the precise pathways are not yet understood. For instance, while 5-HT_{2A} is highly implicated, one study found that ketanserin did not block anti-anhedonic behavioural outcomes and strengthening of excitatory synapses in the hippocampus in mice (Hesselgrave et al., 2021). Thus, while 5-HT_{2A} does likely play a large role in the perceptual changes and the changes in neural plasticity, it is important to systematically test the involvement of each implicated pathway.

There is also the possibility of non-linearity in this pathway. Calder et al. (2023) suggests that one possibility is that AMPA receptor stimulation triggers a feedback loop, in which BDNF is secreted, which stimulates TrkB receptors, which then stimulates mTOR, which once again stimulates further BDNF production and sustained AMPA activation. Given the requirement for mTOR and AMPA activation to be sustained for enhanced neuroplasticity (Ly et al., 2018), this is perhaps a sensible hypothesis. Again, the complexity of this calls for systematic hypothesis testing.

What is also not fully understood is why these structural and functional changes induce therapeutic benefits. In a range of psychiatric disorders, the inverse of the

neurological effects of psilocybin are often seen. For instance, hippocampal atrophy is a common symptom of depression. The prefrontal cortex is also highly implicated in depression, with reduced neuroplasticity and synapse atrophy being seen (Calder et al., 2023). The prefrontal cortex is also implicated in the pathology of anxiety, with deficits in synapses leading to the amygdala (Calder et al., 2023). In fear conditioned mice, psilocybin restores the loss of neurogenesis in the hippocampus (Du et al., 2023). Significant dendritogenesis and synaptogenesis are also seen in the PFC (Calder et al., 2023). The hippocampus is thought to play a role in mood, stress, and cognitive flexibility, and so it is possible that by restoring the neural plasticity in this area we are restoring the deficit, and therefore the dysfunction, caused by depression. Similarly, the plasticity changes in the PFC may allow for greater regulation of the limbic system.

However, there is still further complexity in understanding these outcomes. While psilocybin does attenuate fear conditioning in mice (Zhao et al., 2024), it can also speed its acquisition. Indeed, typical antidepressants such as SSRIs, which are similarly thought to have therapeutic benefits due to increases in neural plasticity, also appear to have a bidirectional behavioural effect. One study showed that mice with a depressive like phenotype showed more depressive behaviour when treated with fluoxetine in a stressful environment, as compared with controls. The environmental interactions of psilocybin in humans have so far focussed on the acute phase of the drug, and indeed, negative psychedelic experiences can lead to decreases in sense of well-being (Calder et al., 2023), and the duration of this negative experience appears to predict greater deficits in well-being (Calder et al., 2023).

As can be seen, while there are many well documented effects of psilocybin, we have yet to fully understand psilocybin's molecular signalling pathway, neural plasticity effects, network changes, and therapeutic outcomes. Furthermore, human and rodent studies are highly limited by cost and time constraints, limiting the rate and extent of discovery.

Planarian flatworms may prove useful in bridging some of these obstacles. Planarian flatworms are a simple animal with surprising analogues to human biology. They possess a body plan common to all mammals, with a dorso-ventral axis, bilateral symmetry, and a central nervous system (Sarnat & Netsky, 1985), (Buttarelli et al., 2008). Planarians possess a bilobed inverted "u" shape brain (Inoue et al., 2017), with commissural fibres connecting the two lobes (Inoue et al., 2017), (Buttarelli et al., 2008). From this brain, two longitudinal nerve cords are placed laterally, and extend down the long axis of the animal (Inoue et al., 2017), (Buttarelli et al., 2008), (Sarnat & Netsky, 1985). Planarians reach adulthood quickly, and are biologically immortal, with pluripotent stem cells called "neoblasts" allowing them to regenerate entire organisms from small body fragments (Newmark & Alvarado., 2000). Planarians also lack a blood-brain barrier, allowing easy drug administration. Their ease of manipulation and relative simplicity while possessing many fundamental

similarities with more complex models makes them a potentially useful supplement to rodent and human studies.

While planarians are simpler than rodents and humans, they are capable of complex behaviour, using multiple sensory inputs (Inoue et al., 2015). These include phototaxis, chemotaxis, thermotaxis, and thigmotaxis (Inoue et al., 2015). Planarians are also capable of memory and learning, as exhibited in both rewarding and punishing behavioural paradigms (Inoue et al., 2017), (Buttarelli et al., 2008). Several drugs of abuse have been investigated in planarians, and withdrawal-like symptoms, behavioural sensitization, and motility changes have all been observed (Buttarelli et al., 2008). Furthermore, SSRIs attenuate fear conditioning in planarians (Cho et al., 2019). This complex but stereotyped behavioural profile indicates that planarians may be useful for testing the effects of psilocybin at the behavioural level as well as the neurological.

The ability of planarians to integrate multiple sensory inputs into their decision making implies the existence of complex neural networks within the animal (Inoue et al., 2017). Indeed, the planarian brain possesses distinct regionalisation (Inoue et al., 2017), (Cebria et al., 2002), with several axon bundles (Cebria et al., 2002). Planarians also possess the synaptotagmin gene homologue DjSYT, which is expressed in neurons throughout the planarian nervous system (Cebria et al., 2002). DjSYT protein structure includes a transmembrane domain and two C2 domains, implying that the fundamental mechanisms by which synaptic transmission occurs in planarians is similar to that in rodents and humans (Cebria et al., 2002). Inoue et al. (2015) found that different sensory input supported decision making by planarians is associated with activity in different regions of the brain, as well as by different neuron populations. Planarians exhibit multipolar neurons, as well as dendritic spines, neuropil and interneurons (Buttarelli et al., 2008), (Sarnat et al., 1985).

These findings imply both the existence of functional neural networks within planarians, and that these neural networks may be mechanistically homologous enough to serve as a generalisable model for the network effects of psilocybin. Given that enhanced dendritic spine growth and thickness is associated with both psilocybin administration and its therapeutic effects (Ly et al., 2018), (Du et al., 2023), planarians may be a useful model to study this specific outcome. Furthermore, the existence of neural networks, and the possibility that these networks are caused by fundamentally similar mechanisms as they are in humans and rodent models, means that planarians may serve as a useful tool for furthering our understanding of the functional connectivity changes induced by psilocybin.

Research has confirmed the existence of several neurotransmitters in planarians. This includes serotonin, dopamine, norepinephrine, epinephrine, GABA, acetylcholine, glutamate, and aspartate (Buttarelli et al., 2008). These neurotransmitters also serve similar functions in planarians as they do in humans. Dopamine stimulates locomotion and stereotyped behaviours, acetylcholine inhibits

locomotion, serotonin regulates circadian rhythms, while GABA and benzodiazepines produce physical dependence (Buttarelli et al., 2008). This translatability is important when considering planarian viability as a model for psilocybin.

There is also evidence of interactions between these neurotransmitter systems, with opioid effects being partially mediated through dopamine systems, dopamine and acetylcholine having opposing effects on locomotion, and cannabinoid effects involving opioid and dopamine systems (Buttarelli et al., 2008). Given the interactions between glutamate, serotonin, and GABA that are thought to underlie psilocybin's plasticity effects (Banks et al., 2021), these stereotyped interactions are of particularly important note.

Research has also identified planarian homologs of various neurotransmitter receptors. This includes serotonin like g-protein coupled receptors (Saitoh et al., 1997), (Saitoh et al., 1996), which LSD was found to bind to and was displaced by serotonin and 5-HT antagonists (Saitoh et al., 1996). These receptors show homology with human serotonin receptors, with the highest similarity being with the human 5-HT_{1A} receptor, which showed 58% identity within transmembrane domains (Saitoh et al., 1996). However, planarian serotonin receptor similarity to 5-HT_{2a} is yet to be reported.

AMPA and NMDA like receptors are also implied to exist in planarians. AMPA and NMDA both induce stereotyped seizure like activity, with this effect being attenuated by topiramate (Rawls et al., 2009). NMDA induced seizure like activity was antagonized by NMDA and AMPA receptor antagonists (Rawls et al., 2009). Metabotropic glutamate receptors have not been directly identified in planarians; however, gene analyses have revealed 9 mGluR-like sequences in the genome of *Schmidtea mediterranea* (Zamanian et al., 2011).

As such, it is possible that the proposed molecular signalling pathways for the neural plasticity effects of psilocybin exist in planarians. For instance, mTOR in planarians is known to play a key role in cell proliferation after amputation (Gonzalez-Estevez et al., 2012). However, planarian neurobiology is not well characterised, and the extent to which homologous pathways exist and control outcomes of interest such as neural plasticity and therapeutic effect is yet unknown.

While planarians prove to be a promising potential model for studying the neuroplastic effects of psilocybin, this is yet to be directly investigated. Our lab has also only started research on planarians this year. As such, there is a need to develop and adapt methods which can investigate these changes in the planarian brain. The present study focuses on developing a method for anti-phospho-histone-H3 (PH3) staining. During cell division, histone H3 becomes phosphorylated at serine 10. Anti-phospho-histone-H3 binds to histone H3 only when it has a phosphate group at serine 10, and thus is an effective marker for proliferating cells. As such, it is also an effective marker for neurogenesis and should allow for detection of

possible neurogenic effects in planarians after psilocybin exposure. Based on the neurogenesis seen in rodent models, we expect planarians to show increased neurogenesis after psilocybin exposure. Additionally, we present pilot data for a basic behavioural assay examining planarian movement after exposure to a range of psilocybin doses. In rodents, psilocybin inhibits locomotion (Tylš et al., 2016) and exploratory behaviour (Nichols et al., 2011), and so we expect to see a lack of movement in psilocybin exposed animals compared to controls.

Methods and materials:

Animals:

Freshwater planarians of an unidentified species were used in this study. The planarians were wild caught from a local spring in Makara. They were maintained in their native spring water at a constant temperature of 20°C.

PH3 staining method development:

The immunohistochemistry (IHC) protocol for detecting cell proliferation in planarians was optimized through three iterations. Due to time constraints, optimizations were not entirely systematic in nature, with multiple changes in each iteration. Iterations past iteration one were adapted from Newmark and Alvarado (2000).

Iteration 1:

Samples were fixed for 20 hours in a 1:3 solution of acetic acid to methanol. Rehydration was performed in two steps: first using a 1:1 mixture of methanol and PBS for 15 minutes, followed by 1x PBS for an additional 15 minutes. Samples were then bleached in 6% hydrogen peroxide in methanol for 4 hours. After bleaching, tissues were blocked with 20% dog serum in 1x PBS. Primary antibody (anti-rabbit - PH3) was applied at both 1:250 and 1:500 dilutions with a 2-hour incubation period, followed by four washes over 35 minutes in 1x PBS. Secondary antibody (goat anti-rabbit Alexa 488) was applied at a 1:500 dilution and incubated for 1 hour. Following secondary antibody application, samples were washed four times over 35 minutes in 1x PBS. Samples were then counterstained with DAPI and mounted in an 80% glycerol/PBS solution.

Iteration 2:

Samples were killed and mucus was removed by treating 2% HCl for 1 minute, followed by fixation in Carnoy's solution (1:3:6 ratio of acetic acid:chloroform:ethanol) at 4 degrees Celsius for 2 hours while rocking. The samples were then incubated in methanol for one hour at 4 degrees Celsius while rocking. Rehydration was performed sequentially: first in a 1:1 mixture of methanol and 1x PBS for 15 minutes, then in 1x PBS for an additional 15 minutes. Samples underwent overnight bleaching in 6% hydrogen peroxide in methanol. Tissues were blocked with 20% dog serum prepared in 1x PBS. Primary antibody was applied at both 1:250 and 1:500 dilutions in 20% dog serum and incubated for 16 hours. Following primary antibody

incubation, samples were subjected to extensive washing (6 times in 1x PBS) over an 8-hour period. Secondary antibody (Alexa 488) was applied at a 1:500 dilution in 20% dog serum and incubated for 16 hours. Another round of extensive washing was performed (6 times in 1x PBS over 8 hours). Samples were then counterstained with DAPI and mounted in 80% glycerol/PBS containing 2.5 mg/ml DABCO.

Iteration 3:

Samples were killed and mucus was removed by treating with 2% HCl for 3 minutes, then fixed in Carnoy's solution at 4 degrees Celsius for 2 hours while rocking. Following fixation, samples underwent a one-hour methanol treatment at 4 degrees Celsius while rocking. Rehydration was performed in two steps: first using a 1:1 mixture of methanol and 1x PBS containing 0.2% Tween 20 for 15 minutes, followed by 1x PBS with 0.2% Tween 20 for an additional 15 minutes. Samples were pre-blocked with 10% skim milk prepared in 1x PBS containing 0.2% Tween 20 for one hour at room temperature. Primary antibody was added into the blocking solution at a 1:500 dilution and incubated for 16 hours. Following primary antibody incubation, samples underwent six 1x PBS + 0.2% tween 20 washes over an 8-hour period. Secondary antibody (Alexa 488) was applied at a 1:500 dilution in 10% skim milk and incubated for 16 hours. Another round of six 1x PBS + 0.2% tween 20 washes was performed over 8 hours. Samples were then counterstained with DAPI and mounted in 80% glycerol/PBS containing 2.5 mg/ml DABCO.

Iteration 4:

Samples were killed and mucus was removed by treating with 2% HCl for 3 minutes, then fixed in Carnoy's solution) at 4 degrees Celsius for 2 hours while rocking. Following fixation, samples underwent a one-hour methanol treatment at 4 degrees Celsius while rocking. Samples were then divided into three treatment groups for the bleaching step: 16-hour bleaching: Samples were bleached in 6% H₂O₂ in methanol for 16 hours; 12-hour bleaching: Samples were bleached in 6% H₂O₂ in methanol for 12 hours; No bleaching: Samples proceeded directly to rehydration. Each bleaching treatment group was further divided into two antibody dilution groups (1:250 and 1:500), except for the no bleaching group where only the 1:500 dilution data was collected. Following their respective bleaching treatments (where applicable), rehydration was performed in two steps: first using a 1:1 mixture of methanol and 1x PBS containing 0.2% Tween 20 for 15 minutes, followed by 1x PBS with 0.2% Tween 20 for an additional 15 minutes. Samples were pre-blocked with 10% skim milk prepared in 1x PBS containing 0.2% Tween 20 for one hour at room temperature. Primary antibody was added into the blocking solution at either 1:250 or 1:500 dilution according to treatment group and incubated for 16 hours. Following primary antibody incubation, samples underwent six 1x PBS + 0.2% Tween 20 washes over an 8-hour period. Secondary antibody (Alexa 488) was applied at a 1:500 dilution in 10% skim milk in PBS + 0.2% Tween 20 and incubated for 16 hours. Another round of six 1x PBS + 0.2% Tween 20 washes was performed over 8 hours.

Samples were then counterstained with DAPI and mounted in 80% glycerol/PBS containing 2.5 mg/ml DABCO.

Psilocybin Preparation and administration:

A stock solution was prepared by dissolving psilocybin in deionised water to a concentration of 1000 μ m. This stock was then diluted in spring water to create the various experimental doses.

Dose-Response and Locomotor Assessment:

Multiple doses of psilocybin (1 μ M, 4 μ M, 10 μ M, 20 μ M, 30 μ M, 40 μ M) were tested to determine an appropriate experimental dose. Each dose was tested on a single planarian (n = 7) placed individually in petri dishes containing 3ml of psilocybin solution or spring water. Planarian behavior was recorded using a digital camera for 12 hours post-administration. Locomotion was quantified using Ethovision to analyze the video recordings. 10 μ m was unable to undergo analysis via Ethovision, and as such was removed from the results.

PH3 staining psilocybin exposed planarians

A dose of 20 μ M was selected for PH3 staining procedure. Twelve planarians were randomly assigned to either the psilocybin treatment group (n = 6) or the control group (n = 6). Groups were placed in petri dishes containing either 3ml of psilocybin or spring water. After 24 hours, planarians were killed using 2% HCl and fixed and stained using iteration 3.

Imaging and Quantification

Stained planarians were imaged using an Olympus IX53 inverted microscope with 4x and 10x objectives. Images were captured using cellSens. Cell proliferation was quantified by counting the number of phospho-histone H3-positive cells.

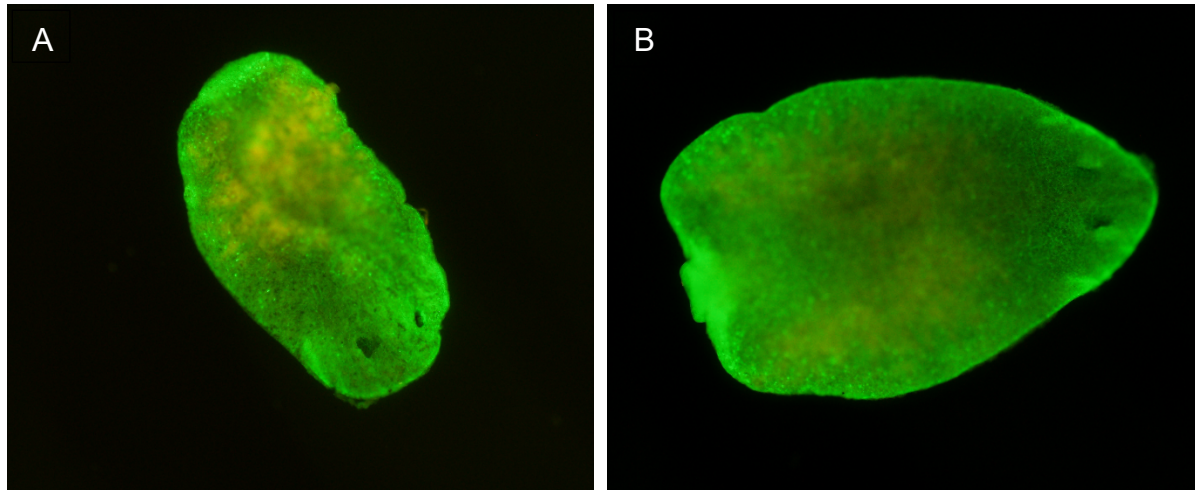
Results

IHC method development

Iteration 1:

Figure 1

Immunofluorescent PH3 staining iteration one



Note. Images of PH3 stained planarians at 4x objective. Panel **A** shows staining at 1:500 primary antibody dilution; Panel **B** shows staining at 1:250 primary antibody dilution. Green fluorescence indicates PH3 stained cells.

Immunofluorescence analysis using PH3 antibody revealed potential specific labelling of proliferating cells. However, specimens exhibited high background fluorescence at both 1:500 and 1:250 primary antibody dilutions (Figure 1A). Significant tissue damage occurred during processing, including detachment of the blastema in one specimen (figure 1B).

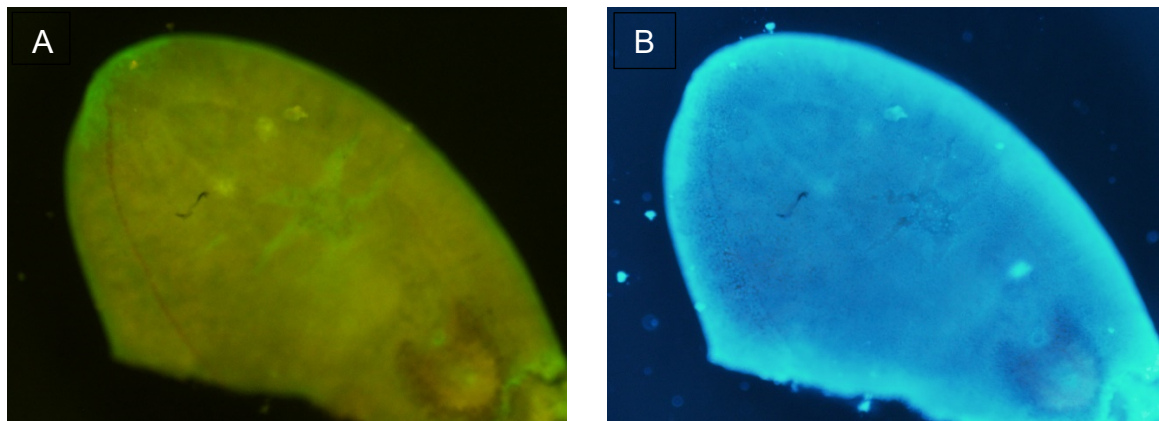
It is possible that the specimens were over fixed, thus leading them to be brittle, damaged, and prone to non-specific staining. Insufficient wash step length may have also increased non-specific staining.

Iteration two:

To improve immunostaining, a previously used method from Newmark and Alvarado (2000) was adapted. This method uses a different fixation method, potentially reducing over fixation. It also includes significant wash steps.

Figure 2

Immunofluorescent PH3 staining iteration two



Note. Images of 1:500 PH3 stained planarian at 10x objective (Sample was damaged during procedure). Panel **A** shows FITC channel Panel **B** shows DAPI channel. Green fluorescence indicates PH3 stained cells.

Implementation of the modified Newmark and Alvarado protocol resulted in minimal specific PH3 immunostaining. Edge-specific fluorescence was observed at 1:500 primary antibody dilution (Figure 2A). DAPI nuclear counterstaining showed diffuse fluorescence without distinct nuclear labelling (Figure 2B).

The edge-specific immunostaining and poor nuclear counterstaining suggest that lack of permeabilization may be an issue with these samples.

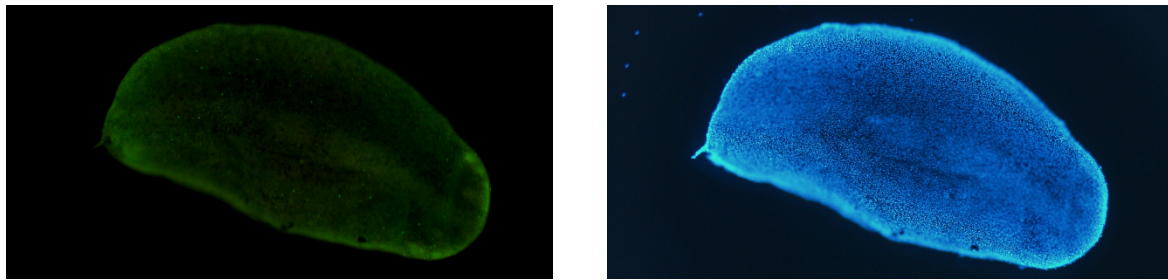
Background fluorescence showed a distinct yellow hue, rather than the consistent Alexa 488 green seen in iteration one. This indicates successful reduction of non-specific staining.

Iteration 3:

Due to potential issues with permeability, HCl 2% exposure time was increased to three minutes. Species and size are known to play a role in mucus output, and thus mucus removal steps such as HCl 2% exposure times need to be adjusted accordingly. Blocking reagent was changed to skim milk and decreased to 10% to allow for greater antibody penetration. 1x PBS was replaced with 1x PBS + 0.2% tween 20 to assist permeability. The 6% h2o2 bleaching step was also removed, as it may have been having negative effects on immunostaining quality

Figure 3

Immunofluorescent PH3 staining iteration three



Note. Images of 1:500 PH3 stained planarian at x4 objective. Panel **A** shows FITC channel. Panel **B** shows DAPI channel. Green fluorescence indicates PH3 stained cells.

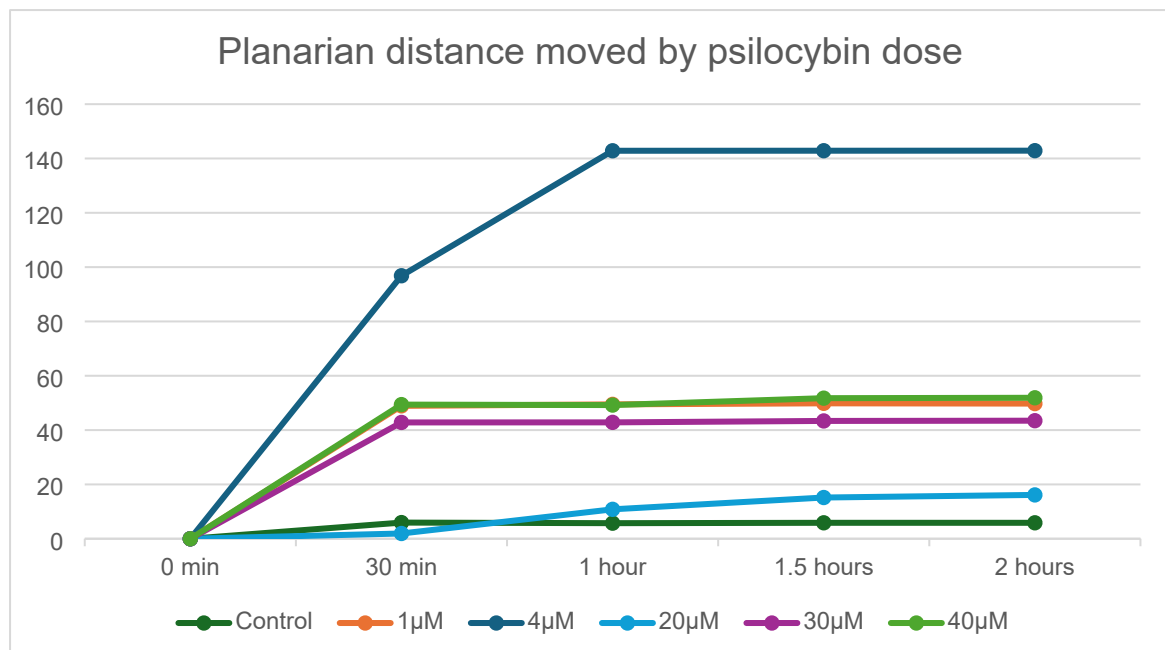
Iteration three showed improved specific immunostaining and signal-to-noise ratio. PH3 positive cells were clearly visible throughout the tissue rather than being limited to the edges (Figure 3A). DAPI nuclear counterstaining showed distinct nuclear labelling (Figure 3B). Background fluorescence was notably reduced compared to previous iterations.

The sample wide PH3 labelling and successful DAPI staining indicate improved permeabilization. Changing 20% dog serum blocking solution to 10% may have assisted in increasing permeability for the anti-PH3. The improved DAPI staining suggests that the increased HCl acid time and addition of 0.2% tween 20 to PBS may have also played an important role in increasing permeability.

Psilocybin

We decided that iteration three's staining was effective enough to use when collecting pilot data about potential neurogenic effects of psilocybin in planarians and moved ahead with the dose-response experiment and staining of psilocybin exposed animals.

Dose-response and locomotion:

Figure 4

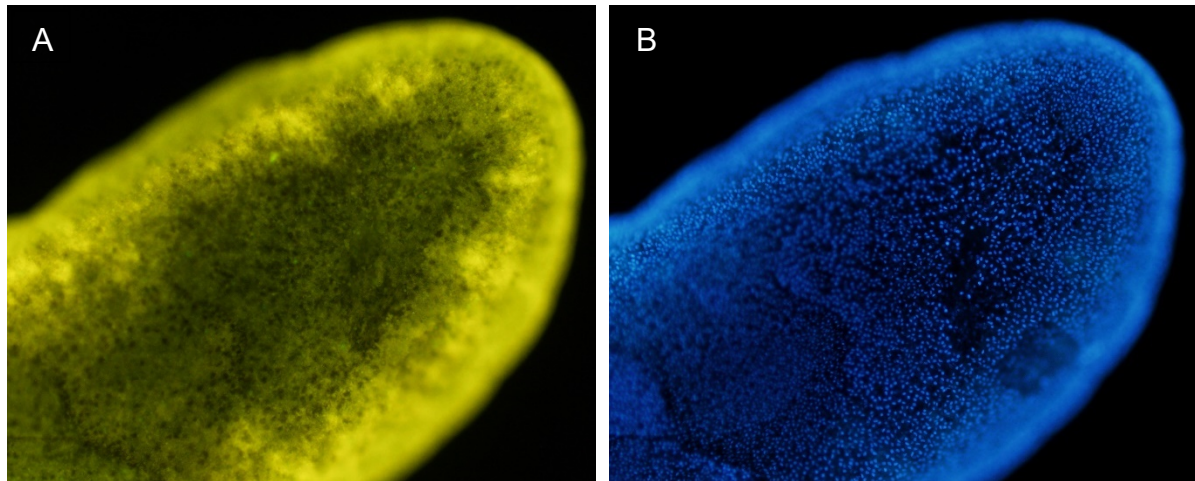
Note. Distance moved in individual petri dishes when exposed to psilocybin.

Planarians show a range of motility. The control planarian showed the least motility, barely moving over the course of the experiment. 1 µM, 30 µM, and 40 µM exposed planarians all showed significant motility within the first 30 minutes but plateaued after this and showed limited movement for the remainder of the experiment. The 4 µM exposed planarian showed a much higher level of movement than all other planarians but this also plateaued after the first hour. The 20 µM planarian was the only planarian with comparable movement to the control planarian, with only slightly more total movement.

PH3 staining psilocybin exposed planarians:

Figure 5

Immunofluorescent PH3 staining of psilocybin exposed animals



Note. Images of 1:500 primary PH3 stained psilocybin exposed planarian at x10 objective. Panel **A** shows sample in FITC channel. Panel **B** shows DAPI channel.

Psilocybin exposed planarians and controls stained using iteration three showed poor immunostaining. Some faint indicators of successful PH3 immunostaining can be seen, however the signal-to-noise ratio is poor. DAPI staining remained effective.

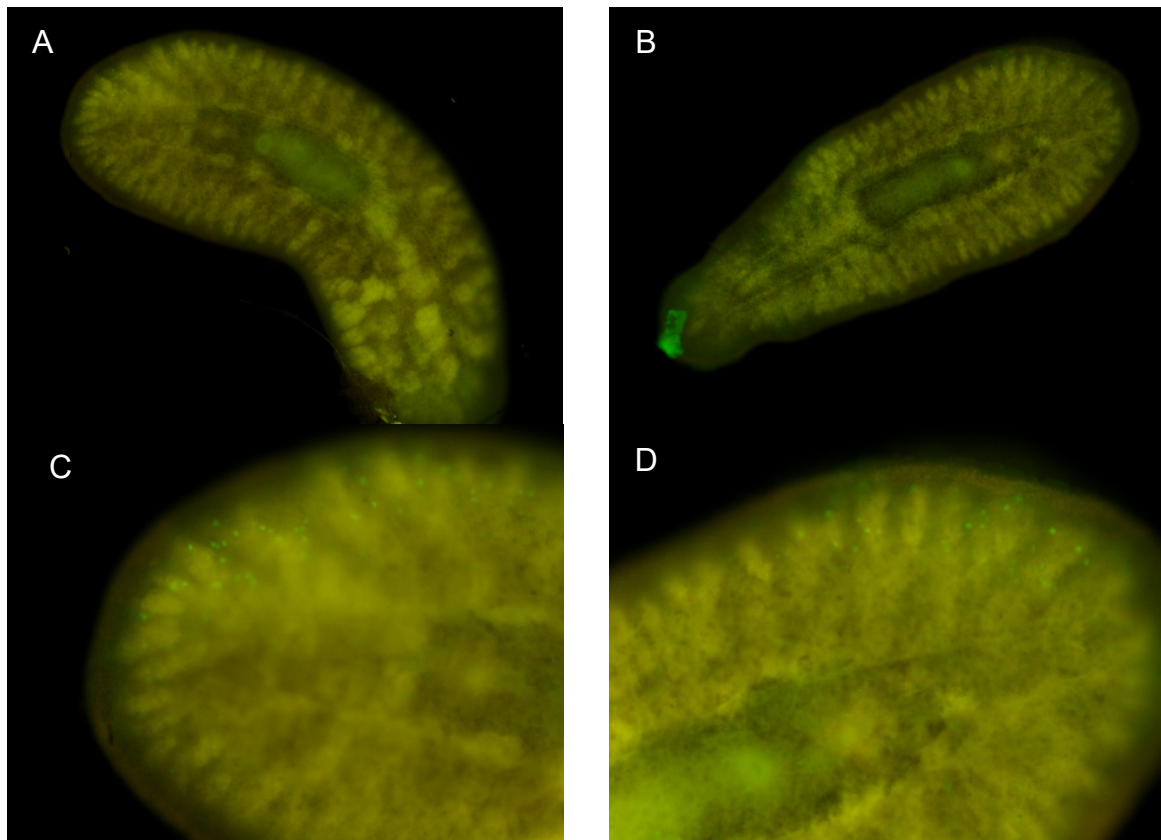
IHC method development revisited

Iteration 4:

Because iteration three showed poor results when reused for specimens in the psilocybin experiments, systematic reinvestigation of the procedure was prompted. The h2o2 bleaching step was reintroduced to investigate the inconsistent background results seen in iteration three and the psilocybin experimental animal immunostaining. Planarians were split into a 16-hour bleach procedure, a 12-hour bleach procedure and a non-bleach procedure. Furthermore, the 16-hour bleach procedure and 12-hour bleach procedure planarians were split into 1:500 and 1:250 primary antibody dilutions. The non-bleached planarian was also split into 1:500 and 1:250 antibody dilutions, but the 1:250 antibody dilution data was lost.

Figure 6

Immunofluorescent PH3 staining iteration four – 16-hour bleach procedure

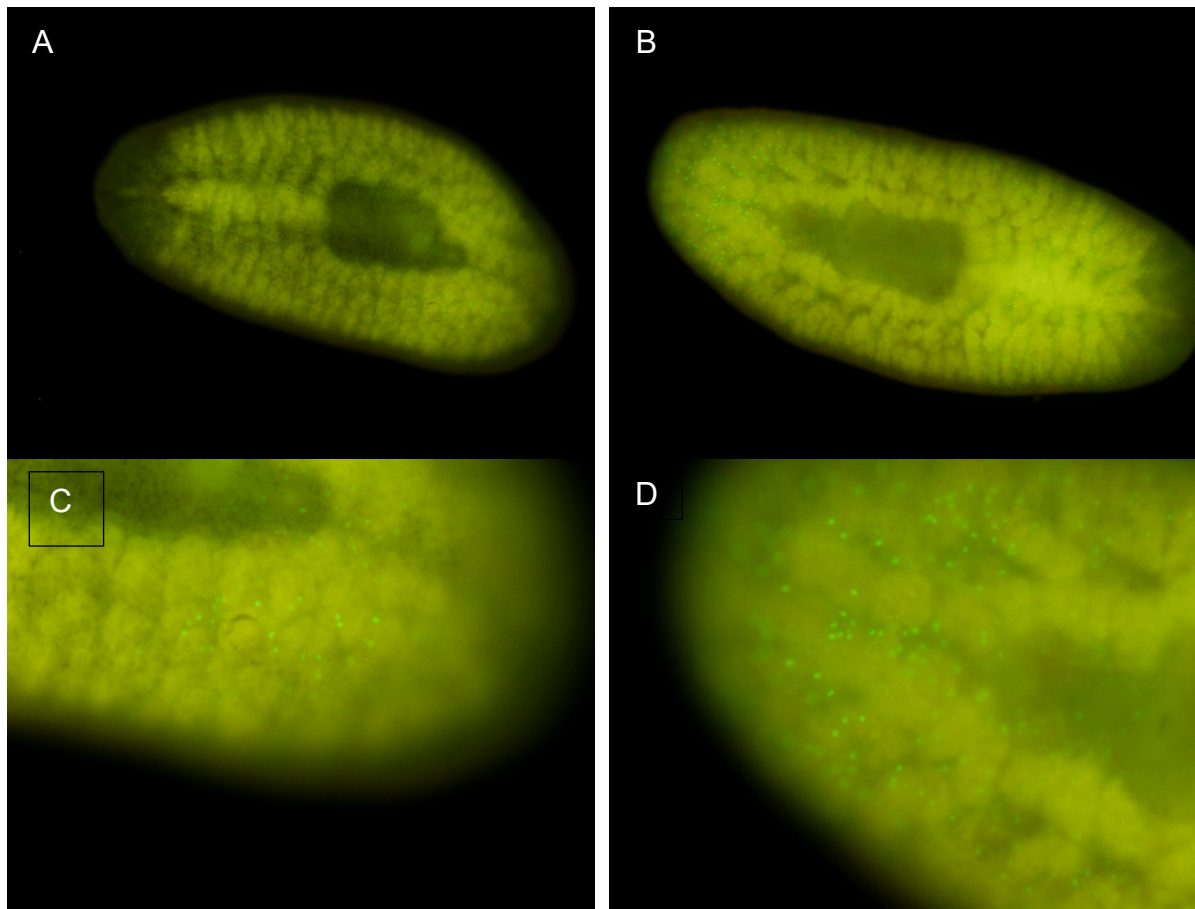


Note. Images of PH3 stained planarians. Panels A and B show samples bleached with 6% H_2O_2 for 16 hours at 1:250 and 1:500 primary antibody dilutions respectively at 4x objective. Panels C and D show 10x objective of A and B, respectively.

Specimens treated with H_2O_2 for 16 hours showed minimal PH3 immunostaining at both antibody dilutions. Specific labelling was primarily confined to the tail region (Figure 6). Tissue pigmentation was effectively reduced, allowing visualisation of internal structures.

Figure 7:

Immunofluorescent PH3 staining iteration four – 12-hour bleach procedure

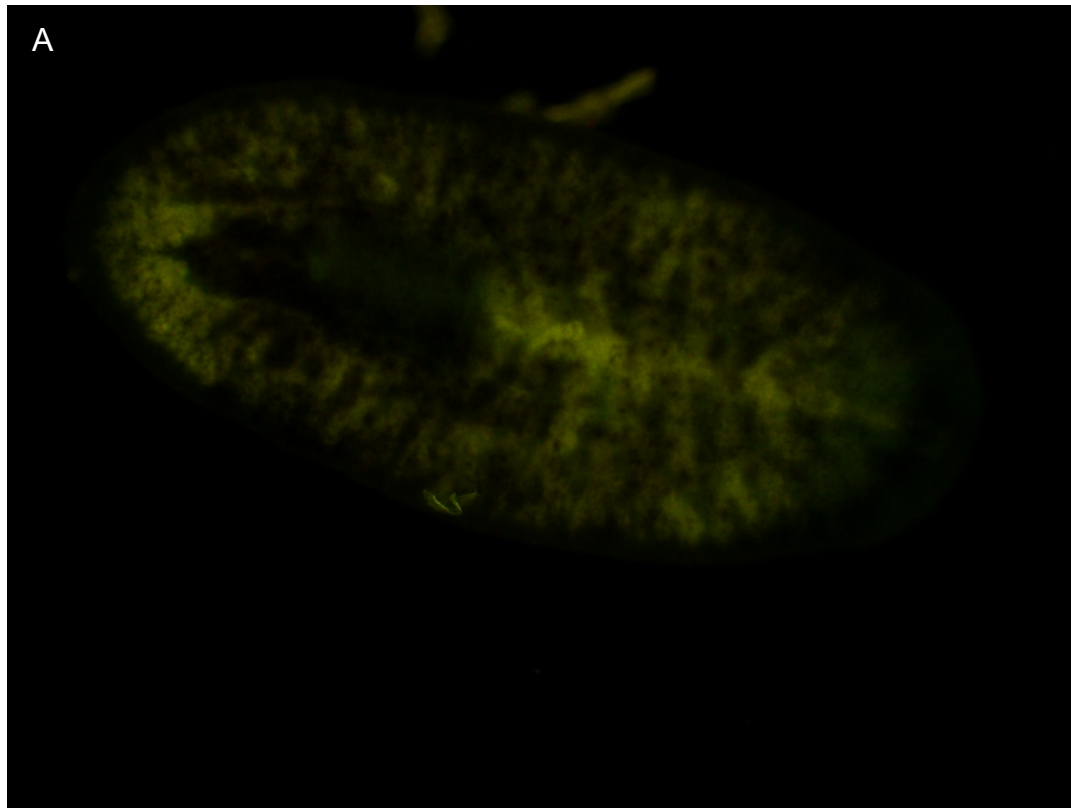


Note. Images of PH3 stained planarians. Panels A and B show samples bleached with 6% H_2O_2 for 12 hours at 1:250 and 1:500 primary antibody dilutions respectively, at 4x objective. Panels C and D show A and B at 10x objective, respectively.

Immunostaining was poor for the 12-hour bleached 1:250 primary antibody dilution, with PH3 staining being limited to the tail region (Figure 7A, C). However, the 1:500 primary antibody planarian showed significant staining throughout (Figure 8B, D). Background appears to be slightly higher than the 16-hour bleached planarians. Pigment removal was similar, as was visualisation of internal structures.

Figure 8:

Immunofluorescent PH3 staining iteration four – no bleach procedure



Note. Images of PH3 stained planarians. Panel A shows sample without h2o2 bleaching at 1:500 primary antibody dilution at 4x objective.

Specimens processed without h2o2 bleaching (1:500 antibody dilution) showed minimal PH3 immunostaining. While background fluorescence was reduced compared to bleached specimens, internal structure visualisation was also reduced.

Iteration 4 summary:

While bleach reduces pigment and allows for better visualisation of internal structures than no bleaching step, it appears that leaving planarians unbleached can result in lower background. However, it also seems that bleaching may play a role outside of pigment, with significant immunostaining seen in the 1:500 antibody dilution for the 12-hour bleached planarian, despite higher background than the nonbleached planarian. This may be a dose dependent effect however, with reduced immunostaining in the 16-hour bleached planarian.

Unexpectedly, the higher antibody dilution appears to result in less effective immunostaining, but not through noticeable increases in non-specific staining. Rather, the 1:250 antibody staining appears limited to the tail regions, while 1:500 antibody staining in the 12-hour bleached planarian was seen throughout the sample.

General discussion

Due to poor staining results of animals exposed to psilocybin and controls, we have failed to produce quantifiable results of cell proliferation after psilocybin exposure in planarians. However, our method development process has produced a variety of results which indicate several key factors underlying successful immunostaining.

Mucus removal appears to be an important step to optimize, given the discrepancy in PH3 and DAPI staining between iteration two and later iterations. Initial 2% HCl time of 1 minute proved ineffective, but increasing this to 3 minutes allowed for improved staining. This is in line with previous literature, which suggests treatment as long as 5 minutes may be necessary for animals of large size. Follow up studies should test the effects of different lengths of HCl treatment, as well as different methods of mucus removal that have been identified within the literature. For example, N-acetylcysteine treatment. This is of particular importance as while too short a treatment may yield insufficient permeability, too long a treatment could damage important structures and negatively affect staining.

Further investigation into the effects of H_2O_2 bleaching is also worth undertaking. It appears that H_2O_2 bleaching may enhance background fluorescence. It may also affect staining in some capacity, with increased staining seen after 12 hours of bleaching. Bleaching may improve staining through increasing permeability, as is suggested in previous literature (Forsthoefel et al., 2014). However, excessive bleaching appears to have detrimental effects on staining. This has also been seen in previous literature (Forsthoefel et al., 2014). Follow up studies should focus on determining the optimal bleaching time and strength for maximising the signalling and visualisation benefits, while minimising the background effects and signal reduction.

A minimal range of antibody dilutions were tested in the present study, but the lack of staining with the 1:250 primary antibody dilution in iteration four suggests further investigation is warranted. Higher antibody dilutions may reduce staining through mechanisms such as steric hindrance. Investigating dilutions lower than 1:500 may show that a 1:500 dilution is also decreasing binding to some extent, although iteration four did show extensive signalling, so this is perhaps unlikely.

The non-bleached planarian results are particularly difficult to interpret. Three samples were tested using the same method, yet different results were seen each time. While effective staining was seen in iteration three, the stained samples from the psilocybin experiment and iteration four showed poor staining. Furthermore, background profiles were potentially similar between iteration three and iteration four, but controls and experimental animals from the psilocybin experiment showed much larger background fluorescence. These findings point to a broader limitation with our study, which is that while within experiments and iterations, animals were homogenous, between experiments they were not. The planarian lab is an ongoing project and is not fully set up, so several batches of wild caught planarians were

brought in. It is possible that these various groups of planarians had unique features which led to the seemingly contradictory results seen. Planarian habitats have also been changed several times between experiments, again possibly influencing staining results.

The planarian dose response curve shows several interesting features, with an apparent increase in locomotion after psilocybin exposure. However, this dose-response curve was not linear, with a lower dose like 4 μ m inducing by far the highest level of movement, while 20 μ m barely induced more movement than the control planarian. This contradicts the findings seen in rodent models; however, it is important to interpret these results with caution due to the small sample size. A high power follow up should be done to verify these results.

Limitations:

The present study has several limitations. The statistical power for both the method development and psilocybin experiments is extremely low, with often only one planarian being used. As discussed earlier, these planarians come from several different batches, and are of an unknown species, reducing the generalisability of our results within our study and compared to other literature.

The non-systematic approach to parts of the method development also limits our ability to establish causal relationships between method changes and outcomes. For instance, we have suggested that increasing the length of the HCl step may have aided staining via improved mucus removal. This may be the case, but tween 20 and skim milk blocking were added in the same iteration, and so we cannot truly disentangle their effects from the increased HCl treatment.

While the fluorescence microscope used in the study provides superficial feedback on method developments, a confocal microscope would provide a means to verify improvements between given iterations. It is perhaps not needed when differences between iterations are so distinct, but given the potential for background interference on such a comparatively large sample like a planarian, confocal imagery is necessary for quantification and therefore verification of further method improvements. Confocal imagery will also be important for quantitative analysis of cell proliferation counts.

Finally, I am inexperienced in molecular biology, so discrepant results may be partially due to researcher error.

Future research:

Once the methods have been optimised, and if increased cell proliferation is demonstrated, an obvious short-term follow up study would be to focus on establishing a dose-response relationship for cell proliferation and investigating the effects of psilocybin on cell proliferation over time. Given the high throughput nature of planarians, testing a wide range of doses with a large sample should be relatively

simple, and will provide valuable information on the dose-dependency of the cell proliferative effects of psilocybin. Similarly, examining the effects at multiple time points will provide valuable information on both the acute and sustained effects of psilocybin, informing future directions and providing valuable comparison to rodent studies on neurogenic effects over time.

While PH3 staining is typically effective and reliable, it stains all proliferative cells. This is problematic for confirmed the detection of neurogenesis. Thus, adapting methods using neuronal markers is of key importance. Dual labelling of BrdU and anti-DjSYT may be useful for quantifying mature neuronal numbers after psilocybin exposure. Fluorescent in situ hybridisation can also be used to mark specific neuronal populations, again in combination with BrdU.

While these studies are undergoing, we should also look to perform genetic sequencing of the planarians used in this study. As mentioned, our planarians are wild caught, and the species is unknown. Furthermore, our planarians vary in terms of size and colour, indicating the possibility of multiple species. Obtaining the genome and identifying the species of our planarians will help situate our findings in the broader literature, while also providing us an avenue to identify homologs of genes thought to play a role in psilocybin's effects. Genetic sequencing is also an important step for developing species-specific primers for PCR, probes for in situ hybridisation, and targets for RNAi experiments. These tools will be essential for both the investigation of psilocybin's effects in planarians and discovering the extent to which they are translatable to rodent models and human studies.

We recommend that future research also identify and investigate the molecular pathways underlying psilocybin's effects in planarians. RNAi and pharmacological antagonists are both common in planarian research, and will likely be of use here. Planarians' lack of a blood brain barrier make pharmacological antagonists particularly valuable for rapid investigation of the molecular signalling pathways. Identifying the extent to which proteins and genes of interest are conserved in planarians would also be informative. For example, BDNF and drebin identification would be useful for both identifying homologous pathways in planarians and quantifying gene expression after psilocybin exposure.

Further useful IHC techniques should be tested. Approaches that can detect changes in neural activity, as well as structural changes at the level of the synapses and dendrites should be prioritised. Dil and electron microscopy may be useful approaches for this, allowing for visual examination of structural changes after psilocybin exposure.

Our locomotion results, which contradict those seen in rodent models, prompts further investigation into the key behavioural features in planarians after psilocybin exposure. Planarian behaviours such as thigmotaxis, chemotaxis, phototaxis, and theromotaxis are well characterised, so paradigms could easily be adapted to investigate the possible perceptual changes psilocybin may induce. Seeing how it

affects other behaviours such as operant conditioning, fear conditioning, and conditioned place preference using drugs would also be of interest.

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